PRE-ALGEBRA (8TH) · UNIT 2: EQUATIONS AND EXPRESSIONS

Phone Plan Showdown

Write and solve equations to find the best cell phone plan for any customer.

Honors

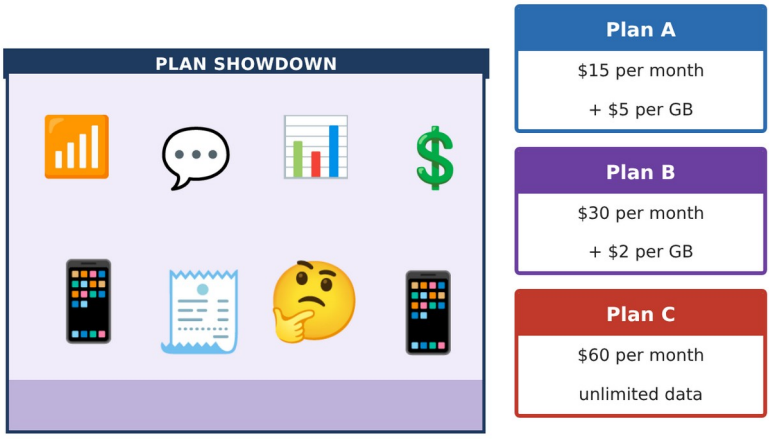
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

A phone store has hired you to help customers choose a plan. Each plan has a monthly fee and a price per gigabyte of data, and customers want to know which plan is cheapest for them.

You will write expressions and solve equations to compare the plans and build a guide that any customer can use.



Learning Goals

- I can model tiered pricing with expressions.
- I can solve and analyze multi-step equations, including equations with parameters.
- I can design a plan and prove when it beats another plan.

Project Timeline

When	What you will do
Day 1	Tiered Plans
Day 2	Equation Analysis
Day 3	Design a Plan
Final	Plan Comparison Guide + reflection

Part 1: Tiered Plans

Day 1 · Plan D: $10 per month plus $6 per GB for the first 5 GB and $3 per GB after that.

1. Write a simplified expression for the cost of Plan D when $g > 5$.

2. Write an expression for the average cost per GB on Plan B.

3. Write and simplify an expression for how much more Plan A costs than Plan B over a year for g GB each month.

Part 2: Equation Analysis

Day 2 · Show each step.

4. When do Plans D and B cost the same? Check that your answer fits the condition $g \geq 5$.

5. Solve $\frac{1}{3}(6g - 9) + 2 = 2(g - 1) + 1$. What kind of solution set is it?

6. For what value of k does $4g + k = 4g + 12$ have infinitely many solutions? For which values does it have no solution?

Part 3: Design a Plan

Day 3 · Create a plan that competes with Plan B.

7. For how many GB is the average cost per GB on Plan B exactly $5?

8. Design a plan that costs more than Plan A at 2 GB but less than Plan B from 8 GB to 12 GB. Show the costs at 2, 8, and 12 GB.

9. Use algebra to prove whether your plan is cheaper than Plan B for every amount of data.

Final Product: Plan Comparison Guide

Your final product must include:

- Tiered-plan expression and average-cost expression
- Equation solutions with solution-type analysis
- Parameter analysis (values of k)
- Your designed plan with a cost table
- A written proof comparing your plan with Plan B

Reflection

R1. Why does the coefficient of g decide whether two plans can ever cost the same?

R2. How is the average cost per GB different from the price per GB?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Expressions and equations model the plans correctly, solutions are correct, and solution types are explained. Goes beyond the minimum with a creative, realistic choice.	Expressions and equations model the plans correctly, solutions are correct, and solution types are explained.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Customer guide clearly states which plan is best for each amount of data and why. Polished, organized, and visually engaging.	Customer guide clearly states which plan is best for each amount of data and why. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.